Retrieval and De-duplication

Retrieval and De-duplication I

Retrieval and De-duplication II

Retrieval dispatches the configured query across 7 independent literature engines (arXiv, OpenAlex, Semantic Scholar, Crossref, PubMed, SovietRxiv, and ChinaRxiv). Each engine is an isolated adapter exposing a uniform `search(query) -> list[Record]` interface; engines that are keyless — arXiv, OpenAlex [@priem2022 openalex], Crossref [@hendricks2020crossref], PubMed/Entrez [@sa yers2022entrez], SovietRxiv / RussiaRxiv, and ChinaRxiv — need no credentials, while Semantic Scholar [@kinney2023semantic] uses a key when present. SovietRxiv is a translated archive of Soviet-era scientific preprints sourced from Math-Net.Ru and CyberLeninka [@sovietrxiv]; ChinaRxiv serves translated Chinese preprints from ChinaXiv via the same unified API. Both retain original-language PDFs alongside each translation, and their polite rate-limit pool (300/min vs 30/min anonymous) is activated by an optional `X-API-Email` header. Optional full-text resolution queries Unpaywall [@piwowar2018state] for open-access locations.

Multiple dispatch degrades gracefully: an engine that is disabled in the configuration, lacks a required key, or cannot reach the network returns a *skipped* status, and the run completes from

Engine Details I

Each engine adapter follows a uniform contract: a module-level API URL constant, a pure `_parse_*` parser function, and a `search_*` entry point with pagination, retry, and graceful error handling. All functions accept an injectable `base_url` parameter for hermetic testing with `pytest-httpserver` — no engine hardcodes its URL inside the function body.

Engine	Rate limit	Pagination	Auth	Records (this run)
arXiv	3s between requests	100/page, offset	Keyless	Sparse
Semantic Scholar	1 req/s (unauth.)	100/page, offset	Optional key	Skipped (429)
OpenAlex	Polite pool (mailto)	200/page, cursor	Keyless	1,000
Crossref	Polite pool (mailto)	1,000/page, offset	Keyless	1,000
PubMed	NCBI usage policy	retstart/retmax	Keyless	986
SovietRxiv	30/min (300/min polite)	1–100/page, cursor	`X-API-Email`	0
ChinaRxiv	30/min (300/min polite)	1–100/page, cursor	`X-API-Email`	0

Engine Details II

SovietRxiv and ChinaRxiv returned zero records for the modafinil query, which is expected: the Soviet-era archive covers mathematics, physics, and engineering preprints, while ChinaXiv covers Chinese scientific preprints, and neither domain has substantial modafinil literature. The engines dispatched correctly, queried the live API, and returned empty result sets without error — confirming graceful degradation.

Canonical Identifier Hierarchy I

Heterogeneous records are reconciled by a **canonical identifier hierarchy** — DOI $>$ arXiv ID $>$ Semantic Scholar ID $>$ OpenAlex ID $>$ a stable digest of the normalized title. When two records share a canonical identifier they are merged, keeping the version with the most complete metadata (a count of non-None optional fields). The DOI is normalized: case-folded, resolver-prefix stripped, so the same paper returned by two engines under case/format-variant DOIs merges. For this run, 2248 records carry DOIs, 932 carry OpenAlex IDs, and 1 carry arXiv IDs. The de-duplicated corpus for this run holds $N = 2302$ records published across 2000–2026.

Relevance Filtering I

After de-duplication, a relevance filter drops papers whose title and abstract contain none of the configured relevance keywords (modafinil, armodafinil, provigil, wakefulness, narcolepsy, cognitive enhancement, alertness, sleep deprivation, vigilance, eugeroic). Keywords are matched case-insensitively; an empty keyword list is treated as no filter to avoid silently wiping the corpus. A year filter then excludes papers published before the configured start year (2000).